

A new method for day-ahead sizing of control reserve in Germany under a 100% renewable energy sources scenario



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ABSTRACT

In Germany, the installed capacity of renewable energy sources, especially that of wind and photovoltaic energy, has increased over the past few years and will continue to increase in the future. Due to errors in forecasting wind and photovoltaic energy, the control reserve needed to balance the electricity system will correspondingly increase if control reserves will be sized statically for several months or one year as it is done in most countries today [1–3]. That is because sizing control reserves this way does not consider the fact that there will be hours with a high penetration of wind and photovoltaic which cause a different demand for control reserves than hours with a lower penetration. Therefore, in this work, we present a new probabilistic *dynamic method* that sizes control reserves for the single hours of the following day making use of forecasts of the power feed-in of wind and photovoltaic. In contrast to similar approaches [2,3] forecast errors of wind and photovoltaic power are not modeled as normal distributions, which does not reflect reality [4–6], but by kernel density estimation to get more realistic distributions. Under a 100% renewable energy scenario for Germany, the control reserve that would be allocated by the *dynamic method* is compared with the control reserve that would be allocated by a *static method*. The *static method* is similar to the probabilistic *Graf-Haubrich method*, which is applied in Germany today, but can, in contrast to this method, be applied to future scenarios. It is shown that the *dynamic method* halves the average required control reserve.

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1. Introduction

To guarantee the secure supply with electrical energy load and generation in a power system always have to be in equilibrium. To compensate for imbalances the transmission system operators connected to the UCTE network can perform control actions with different characteristics and qualities. Therefore they contract different types of control reserves. Primary control reserve must be activated within 30 s. The size of the overall primary control reserve for the UCTE network is set with respect to the reference incident to 3000 MW and then distributed amongst the connected transmission system operator [7]. For this reason the sizing of primary control reserves is not considered in this work. In addition secondary and tertiary control reserves are contracted. In Germany, for instance, these reserves have to be activated within 5 min, respectively 15 min. The sizing of these two reserves is done by each control area individually. Currently, both deterministic and

probabilistic methods are used for the sizing. An example of such deterministic methods is found in the ENTSO-E Handbook [7]:

$$P_{\text{Secondary_reserve}} = \sqrt{a \cdot I_{\text{max}} + b^2} - b \quad (1)$$

Here the amount of secondary control reserve is calculated based on the maximum load I_{max} and two constants a and b , which are determined empirically.

In contrast to deterministic methods, probabilistic methods try to reflect stochastic system behavior [8], ensuring that the risk of an insufficient amount of control reserve is known. One example of a probabilistic method is the so-called *Graf-Haubrich method*, which is used in Germany [9,10] and basically explained in [11]. The method is used to size secondary and tertiary control reserves every quarter for the following three months, without considering subsequent renewable energy sources forecasts.

The *Graf-Haubrich method* is based on the idea that there are different error types that require control reserve. The error types are described in detail in Section 2. For each error type, a distribution is estimated based on historical data, and these error distributions are convolved to produce two summed-error distributions, one for the secondary control reserve and one for the total control reserve. The difference between the distributions for the secondary and total

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control reserves is that different error distributions are considered. From these two distributions – and with fixed targets for surplus and deficit probabilities – the required secondary and tertiary control reserves can be obtained. The demanded tertiary control reserve is the difference between the total and the secondary control reserve. In the following, “control reserve” will be used as a generic term for tertiary, secondary and total control reserves.

To determine the influence of increasing wind penetration levels on control reserves one can use different approaches [12,13]. The *Graf-Haubrich* method does not allow the sizing of control reserves for future scenarios with different wind and photovoltaic production capacities because the distribution of the forecast error, which combines the wind, photovoltaic and load forecast errors (s. Section 2), is based on the required total control reserve of the previous year, which is only valid for the installed capacity of wind and photovoltaic energy of this specific year. To enable the *Graf-Haubrich* method to determine the influence of increasing wind and photovoltaic penetration the wind, photovoltaic and load power forecast errors have to be separated so that forecasts for the period for which the control reserve is sized can be used to calculate the error distributions [14]. The separation of these forecast errors is implemented in the *static method* in this work and has already been discussed, inter alia, in [1–3,14,15]. In any case, these studies concluded that the control reserve requirements will increase with the increasing shares of renewable energy sources and that the increase will depend strongly on the quality of their forecasts.

However there will be a very different demand for control reserve in different hours. Hours with lot of wind and sun will for example lead to a higher demand for control reserve than night hours with little wind where no photovoltaic and only a small wind forecast error can be expected. To benefit from this fact the sizing of control reserves has to be done on a daily or even shorter basis so that forecasts for wind and photovoltaic energy can be used. For reducing the lead time a daily tendering of the control reserves is necessary, which is already the case for the tertiary control reserve and is planned for the secondary control reserve in Germany [16]. Another prerequisite for obtaining the full benefit from renewable energy sources forecasts is reducing the control reserve market product length to, for example, 1 h. In [2,3,17,18] methods for sizing control reserves in day-ahead or shorter time-periods are proposed.

In this work we developed a *dynamic method* which is based on the idea of the *Graf-Haubrich* method and characterized by sizing the control reserve one day before its use for a product length of 1 h. In [2,3] similar approaches have been implemented. However both works model the wind and photovoltaic forecast errors as simple normal distributions which does not reflect reality [4–6], especially for single hours. Here is the main difference to the *dynamic method* presented in this work. For the *dynamic method* the distributions of the wind and photovoltaic forecast errors are created with kernel density estimation depending on the day-ahead forecast and the intraday forecast error to reflect the specific characteristics of wind and photovoltaic forecasts.

In Section 2, the *dynamic method* is described and compared in detail with a modification of the *Graf-Haubrich* method here called the *static method*. In Section 3, the 100% renewable energy sources test scenario is presented followed by a discussion of the results in Section 4 and concluding remarks in Section 5.

2. Comparison of the dynamic and static methods

Because the *Graf-Haubrich* method currently used by the German transmission system operators cannot be applied to a 100% renewable energy sources scenario, a new method that can yield similar results is needed; the *static method*, a modification of the

Graf-Haubrich method, is one such method. The *static method* is used as a baseline method against which the *dynamic method* will be compared. There are two differences between the static and the *Graf-Haubrich* method. First, in the former, the forecast error is separated into wind, photovoltaic and load forecast errors. Second, the *Graf-Haubrich* method uses historical data from the last 12 months to size the control reserve for the next three months, which cannot be performed for a future scenario because historical data from the year before the future scenario and from the scenario year itself would be needed. Therefore, the control reserve is sized for the whole scenario year, which is the usual approach when the control reserve is sized for future scenarios [1,14,15].

Both the *static* and the *dynamic method* respect the same error distributions types. The load forecast error considers the deviation of the quarter-hourly average of the load from the forecasted load for this period whereas the load noise error describes the deviations of the load from its quarter-hourly average value. The schedule step error represents deviations from the scheduled exchange with other control zones. Failures of conventional power plants are represented by the power station error distribution. The wind and photovoltaic forecast errors correspond to the deviations between the actual feed-in and the forecasted values averaged over 15 min.

To illustrate the differences between the two methods, the error distributions for the whole year (*static method*) and for one single hour (*dynamic method*) of the 100% renewable energy sources scenario are presented in Fig. 1 as an example. The methods require the same type of error distributions, but the estimation of the error distributions varies among the two because forecasts for the period for which the control reserve is sized are used in the case of the *dynamic method*. The *static method* sizes the control reserve for the whole year. The *dynamic method* sizes the control reserve for every hour of the following day. Thus, in the markets for control reserve, a product length of 1 h and a lead time of one day are assumed. Other lead times are also possible, but because the German control reserve is tendered daily for the tertiary control reserve and will presumably be tendered daily for the secondary control reserve [16], a shorter lead time is not plausible. A product length of 15 min is not considered because the resolution of the scenario is 1 h.

In Fig. 1, the Gaussian distributions for the load forecast error and load noise errors are narrower for the *dynamic method* because the load of that particular hour is used as the expected value, which is smaller than the maximum load of the year used for the *static method*. With the *dynamic method*, the power station error has a probability of one for an error of zero because there is no residual load in this particular hour and so no conventional power stations are in operation. This is the same with the schedule step error because there are no changes in the power exchange to neighboring countries during this particular hour and so no schedule steps. The difference between the wind and photovoltaic energy forecast error distributions for the static and *dynamic method* is observed because for the *static method*, all 1-h forecast errors of the last year are used, and for the *dynamic method*, mainly the 1-h forecast errors with day-ahead forecasts similar to those for this particular hour are taken into account (s. Section 2.5). It can be observed that the shape of the wind forecast error distribution for the *dynamic method* is not as symmetric as the error distribution for the *static method*. This discrepancy occurs because a high power feed-in compared to the installed capacity is forecasted for this hour, and in contrast to the *static method*, the *dynamic method* can model the fact that in this case, a negative forecast error is more likely than a positive forecast error. In addition, the photovoltaic error distribution for the *dynamic method* is narrower than the distribution for the *static method* because the sun's altitude is low at this hour, and therefore, the *dynamic method* models the fact that the actual feed-in can only vary between zero and the theoretically maximal feed-in due to the sun's position.

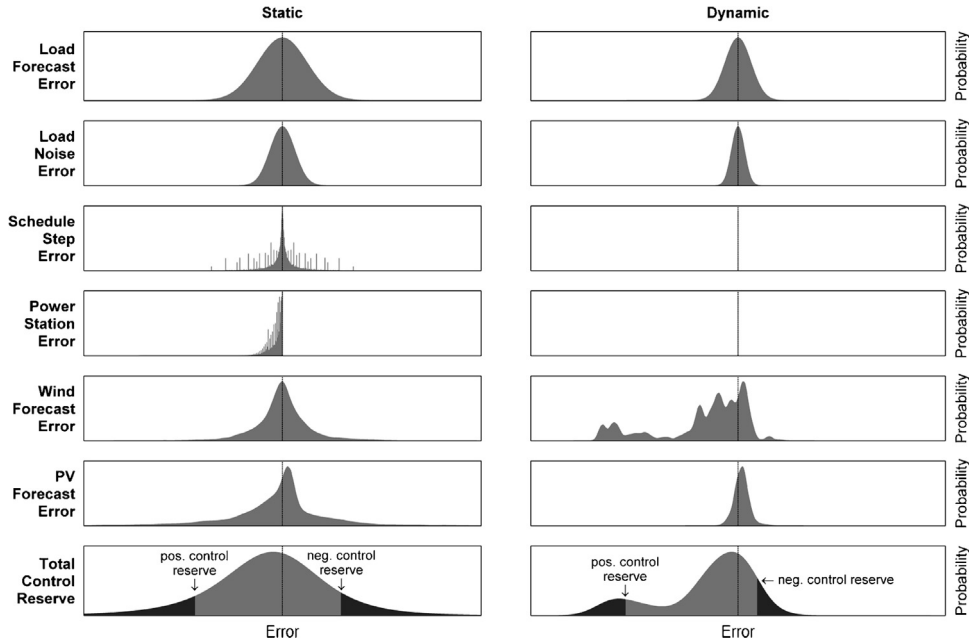


Fig. 1. Illustration of the convolution of the error distributions for calculating the total control reserve distribution and illustration of the determination of the positive and negative total control reserves for a surplus and deficit probability of 10% for the *static* and *dynamic method*.

In the following, the calculation of the different error distributions for the *static* and *dynamic method* is described.

2.1. Load forecast error

With the *dynamic method*, the load forecast for the specific hour considered is needed. With the *static method*, the maximum system's load of the year considered is needed. It is assumed that the load forecast error distribution is a Gaussian distribution [9]. Based on the load value, together with the standard deviation σ and the expected value μ , the error distribution ed can be calculated as follows [19]:

$$ed_{LE}(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \quad (2)$$

where x is the error. The standard deviation depends on the load.

2.2. Load noise error

Because the load noise error is also characterized by a Gaussian distribution [9], the calculation is the same as that for the load error (s. Section 2.1). Only the values for the standard deviation and the expected value are different.

2.3. Schedule step error

Every hour has five schedule steps because electricity is traded in 15-min units (Fig. 2). Therefore, for the *dynamic method*, the forecast of the five schedule steps' values for the specific hour considered is needed. For the *static method*, a time series of all schedule steps for the last year is needed.

The error distribution is calculated based on the ramp model described in [9]. The higher the schedule step is, the wider the error distribution becomes. Fig. 2 shows an example for the five schedule steps within 1 h. The schedule steps at the beginning and at the end of the hour are only considered half of the total number.

2.4. Power station error

For the *static method*, a list of all power stations is needed. For the *dynamic method*, only a list of power stations presumably in operation during the specific hour is needed. The error distribution is calculated in the same way in the two methods. First, the error distribution of one power station is calculated. This error distribution consists of the probability of total failure, partial failure or undisturbed operation (Fig. 3). These values are assigned to the associated power values. Second, the error distributions of all power stations are convolved to create a single error distribution.

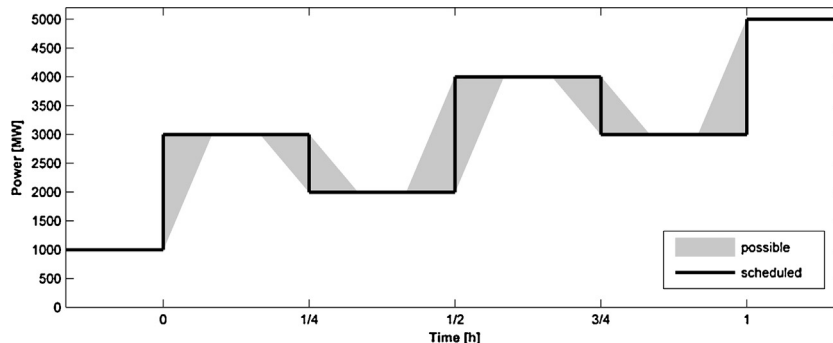


Fig. 2. Planned power exchange with neighboring countries and possible deviations due to the resulting schedule steps for one sample hour.

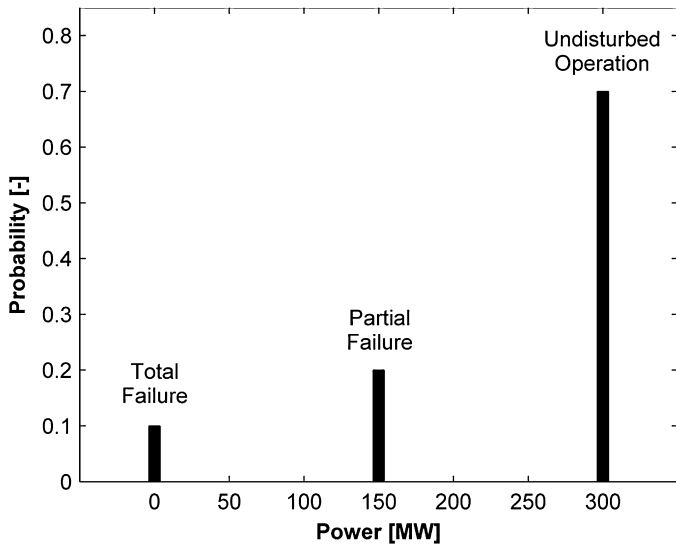


Fig. 3. Example of an error distribution of a power station consisting of two units each with a generation capacity of 150 MW.

2.5. Wind forecast error

An indicator of the quality of forecasts is the normalized root mean square error, which is defined as follows:

$$\text{nRMSE} = \frac{\sqrt{\sum_{i=1}^n (x_{1,i} - x_{2,i})^2 / n}}{x_{\max} - x_{\min}} \quad (3)$$

where x_1 is the forecast value, and x_2 is the real measured value. n is the number of samples, and $x_{\max}$ and $x_{\min}$ are the maximum and minimum values of the sample, respectively.

2.5.1. nRMSE factor

The nRMSE factor is needed if the nRMSE of the historic time series used to create the error distribution for the wind forecast error is different from the nRMSE assumed for the scenario for which the control reserve is sized.

If the nRMSE of the historic time series is greater than the nRMSE of the scenario for which the control reserve is sized, the time series is multiplied by an nRMSE factor.

$$\text{nRMSE factor} = \frac{\text{nRMSE}_{\text{scenario}}}{\text{nRMSE}_{\text{time series}}} \quad (4)$$

The nRMSE factor is the ratio of the nRMSE of the scenario to the nRMSE of the historic time series. The nRMSE of the scenario

might be better because of assumed improvements in the forecast quality in the scenario.

With the *static method*, merely the historic time series of the 1-h forecast error is multiplied by the nRMSE factor because only this historic time series is needed to size the control reserve. Only this series is needed because it is possible to trade electricity up to 45 min before delivery in Germany. With the *dynamic method*, two nRMSE factors exist. In addition to the nRMSE factor for the 1-h forecast error, a factor for the day-ahead forecast is also needed because the control reserve is sized one day ahead and the error distribution of the 1-h forecast error has to be determined based on the day-ahead forecast. In the case of the day-ahead forecast, only the day-ahead forecast error is multiplied by the nRMSE factor. The new day-ahead forecast time series is created by adding the power feed-in to the day-ahead forecast error.

2.5.2. Kernel density estimation

Kernel density estimation is a method for estimating the distribution of a random variable, for example, the wind forecast error. With the *static method*, kernel density estimation is used to calculate the error distribution of the 1-h forecast error. With the *dynamic method*, however, kernel density estimation is used to create an error distribution with two dimensions: one for the 1-h forecast error and one for the day-ahead forecast. In the following, the kernel density estimation procedure for the *dynamic method* is described.

For the *dynamic method*, a historic time series with the day-ahead forecast and the associated 1-h forecast error for the same hour is needed. Every data pair of the 1-h forecast error and day-ahead forecast is replaced by a Gaussian distribution for kernel density estimation. All Gaussian distributions are summed to create one distribution [20]:

$$f(y_1, y_2) = \frac{1}{n} \sum_{i=1}^n \frac{e^{-\frac{1}{2} \left(\frac{y_1 - y_{1,i}}{h_1} \right)^2}}{h_1 \sqrt{2\pi}} \cdot \frac{e^{-\frac{1}{2} \left(\frac{y_2 - y_{2,i}}{h_2} \right)^2}}{h_2 \sqrt{2\pi}} \quad (5)$$

f is the probability of the data pair (y_1, y_2) . y_1 is the 1-h forecast error, y_2 is the day-ahead forecast and n is the number of samples. h_1 and h_2 are the bandwidths for the 1-h forecast error and the day-ahead forecast, respectively. The bandwidth must be greater than 0. For the 1-h forecast error, the bandwidth is calculated as follows [20]:

$$h_1 = \left(\frac{4}{(d+2)n} \right)^{\frac{1}{d+4}} \sigma_1 \quad (6)$$

The bandwidth h_2 for the day-ahead forecast is calculated in the same way. d is the number of dimensions, which is two for the *dynamic method* and one for the *static method*. σ_1 is the standard

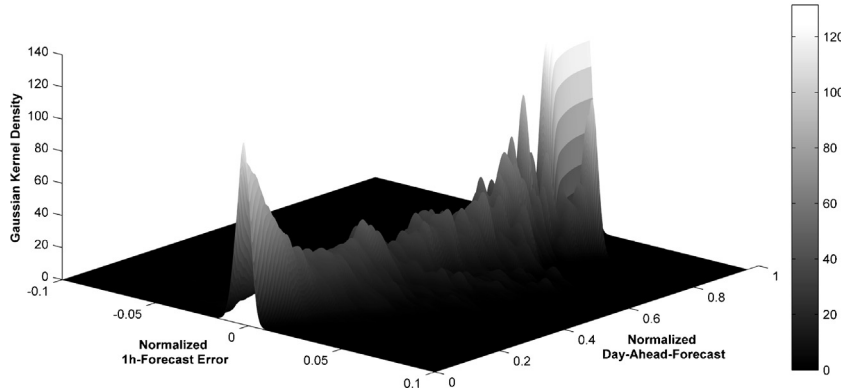


Fig. 4. Two-dimensional kernel density estimation of the wind energy forecast error using the example of a normalized day-ahead forecast and a normalized 1-h forecast error.

deviation for the 1-h forecast error, which must also be calculated for the day-ahead forecast. According to [20], a robust estimate of σ is preferable to the usual sample standard deviation to accommodate long-tailed distributions and to handle possible outliers. Therefore, the median absolute deviation estimator is used, which is calculated as follows for the 1-h forecast error:

$$\sigma_1 = \frac{\text{median}|y_{1,i} - \tilde{y}_1|}{0.6745} \quad (7)$$

$\tilde{y}_1$ is the median of the sample.

Fig. 4 shows an example of a two-dimensional kernel density estimation of the wind energy forecast error.

Finally, the day-ahead forecast for the hour for which the control reserve is sized is used to create a 1-h forecast error distribution. With the help of the day-ahead forecast, the 1-h forecast error distribution is cut out of the kernel density estimation. Because the kernel density estimation is carried out with normalized time series, the error distribution is multiplied by the installed wind energy capacity of the scenario.

In the *static method*, the formula used to calculate the distribution simplifies to

$$f(y_1) = \frac{1}{n} \sum_{i=1}^n \frac{e^{-\frac{1}{2} \left(\frac{y_1 - y_{1,i}}{h_1} \right)^2}}{h_1 \sqrt{2\pi}} \quad (8)$$

The bandwidth and the standard deviation are calculated in the same way as for the *dynamic method*.

2.6. Photovoltaic forecast error

The procedure used to create the photovoltaic forecast error is the same as that used to calculate the wind energy forecast error distribution. The only difference is that the historic time series for the kernel density estimation are not normalized to the installed photovoltaic capacity but to the maximum power output. The maximum power output is limited by the position of the sun. The advantage is that, if for morning hours the maximum power feed-in is predicted, no negative control reserve is sized. If the sun's position is not considered, a negative control reserve would be set because a maximum-power feed-in during the morning hours is tantamount to a middle-power feed-in around midday. Moreover, for a middle-power feed-in around midday, a negative control reserve is needed.

2.7. Determining secondary and tertiary control reserves

The error distributions for the *dynamic* and *static method* are calculated differently but the estimation of the secondary and tertiary control reserves based on the error distributions is performed in the same way. The total control reserve distribution is estimated by convolving all error distributions (s. Fig. 1). The secondary control reserve distribution is also estimated by convolving all error distributions, but in contrast to the total control reserve distribution, scaled error distributions for the load error, the wind energy forecast error and the photovoltaic forecast error are considered. Back-testing the static-method with data from 2010 in [1] resulted in a scaling factor of 0.33 to receive a demand for secondary control reserve comparable to the actually tendered reserves. Assuming X is the original random variable with its distribution function $f_X(x)$ the new random variable Y is:

$$Y = u(x) = 0.33 \cdot X \quad (9)$$

The new distribution function $f_Y(y)$ is now calculated by [21]

$$f_Y(y) = f_X(u^{-1}(y)) \left| \frac{d}{dy} u^{-1}(y) \right| \quad (10)$$

Table 1

Overview of the energy consumption and production of the different producers and consumers in the 100% renewable energy sources scenario adopted from the Kombikraftwerk 2 project [22].

Producer	Energy (TWh)	Consumer	Energy (TWh)
Wind energy	356.4	Load	523.6
Photovoltaics	117.6	Battery	6.7
Bioenergy	34.2	Electrolysis	105.7
Water	25.0	Pump storage	9.3
Geothermal power	39.0	Export	0.0
Battery	5.7	Network losses	12.1
Renewable methane	58.5		
Pump storage	7.0		
Import	13.5		
Total	656.9		657.4

By the described transformation the error distribution the distribution becomes narrower.

The secondary control reserve is sized using the secondary control reserve distribution and the associated surplus and deficit probabilities. With the surplus probability, the amount of negative control reserve is set, and with the deficit probability, the amount of positive control reserve is sized. The tertiary control reserve is sized in two steps. The first step is the calculation of the total and the secondary control reserve, given the appropriate distributions and target probabilities for surplus and deficit. The second step is the calculation of the difference between the total and secondary control reserve, which is the tertiary control reserve.

3. Data

The 100% renewable energy sources scenario was created within the project Kombikraftwerk 2. The derivation of the installed capacities and of the power time series of the different types of consumers, storage and producers is detailed in [22]. Table 1 shows the corresponding energy consumption and production. The resolution of the power time series is 1 h.

For the *dynamic method*, the lead time is one day, which indicates that the control reserve is sized one day ahead. The deficit and surplus probabilities are 0.0225% for the total control reserve and 0.0025% for the secondary control reserve [10]. Consequently, the total security level is 99.95%.

1. Load and load noise error: For the *dynamic method*, the load of the particular hour considered is needed. Because no load forecast has been established for the project, the load of the hour is adopted. For the *static method*, the maximum load of the year is used. The standard deviation of the load forecast error of 0.008% of the annual peak load, and the standard deviation of the load noise error of 0.004% of the annual peak load are taken from [14].
2. Schedule step error: For the *dynamic method*, the five schedule steps of the particular hour considered are needed. Because the resolution of the power time series is 1 h, the 15-min values are created by interpolation. For the *static method*, all schedule steps are used.
3. Power station error: In the scenario considered, combined cycle power plants are used to convert renewable methane to electricity. For the static and dynamic methods, the failure and partial failure probabilities are needed, which can be adopted from [9]. The failure probability is 12.1 per year, and the partial failure probability is 7.3 per year. The total number of control hours is assumed to be 4000 h [23]. It is assumed that secondary control compensates the failure in the first 15 min and then tertiary control overtakes for 45 more minutes because the balance responsible party is responsible for balancing the failure no later than 60 min after the incident.

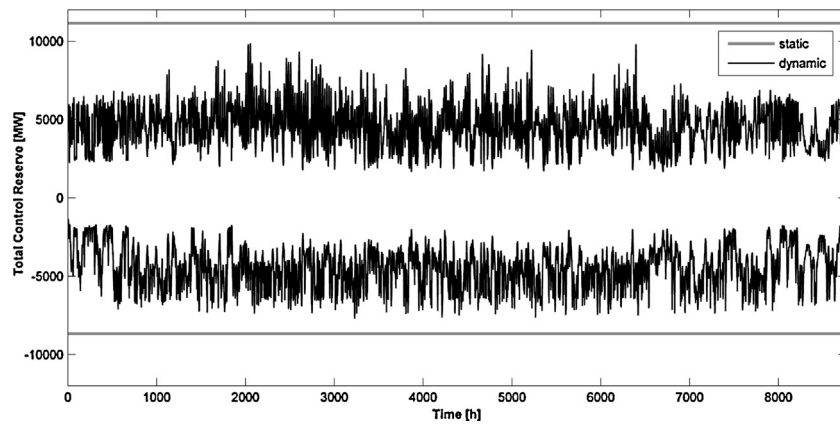


Fig. 5. Positive and negative total control reserve for the 100% renewable energy sources scenario for the *static* and the *dynamic method*.

4. Wind energy forecast error: The kernel density estimation is based on data from 2010. For the *static method*, only a historic time series of the 1-h forecast error is needed, which is the difference between the 1-h forecast, which was established using a model developed at IWES, inter alia, and used in [24], and the power feed-in of the wind energy [25]. For the *dynamic method*, a historic time series of the day-ahead forecast is also needed, which can be downloaded from [25]. Because it can be expected that the forecasts will improve in the future, an nRMSE of 0.78% is assumed for the scenario, representing an improvement of 10% compared to the nRMSE of the dena 2 study [11], in which an nRMSE of 0.86% is predicted for a combined wind energy offshore and onshore forecast error for the year 2020. The improvement is based on a more widespread online monitoring of wind farms compared to the assumptions made in the dena 2 study, which improves the 1-h forecasts. It is assumed that the nRMSE of the day-ahead forecast remains constant because online monitoring does not improve the day-ahead forecast. To size the control reserve one day ahead, the day-ahead forecast is needed. However, instead of the day-ahead forecast, the power feed-in of the wind energy in the 100% renewable energy sources scenario is adopted because both measures have approximately the same average value and show the same fluctuations. Therefore, by adopting the power feed-in, the results are not affected and the required effort is reduced because no day-ahead forecast has to be made for the 100% renewable energy sources scenario.

5. Photovoltaic forecast error: The same data required for the wind energy forecast error are required for the photovoltaic forecast error. For the kernel density estimation, the year 2010 is also considered. The historic time series of the day-ahead forecast and the power feed-in can be downloaded from [25]. The 1-h forecast was calculated using a persistence model that also considers the day-ahead forecast. The model was created as part of a master thesis at Fraunhofer IWES. For the scenario, an nRMSE of 1.01% is assumed, which is 30% greater than the nRMSE of the

wind energy forecast because the photovoltaic forecast is more complex than the wind energy forecast [26]. As for the wind energy forecast, it is assumed that the nRMSE of the day-ahead forecast remains the same. The day-ahead forecast for sizing the control reserve also equals the power feed-in of the 100% renewable energy sources scenario.

4. Results

Fig. 5 shows the positive and negative total control reserves for the *dynamic* and *static method* for the 100% renewable energy sources scenario and for the whole year.

Two interesting results can be observed. First, the total control reserve can be halved in a 100% renewable energy sources scenario if the *dynamic method* is used, even though the security level always equals 99.95%. This finding demonstrates the advantage of considering forecasts in calculating the error distributions. Second, the maximum total control reserve for the *dynamic method* is always smaller than the total control reserve for the *static method*.

A comparison of the control reserve for the 100% renewable energy sources scenario obtained using the *static* and *dynamic methods* with the control reserve in Germany in 2012 shows that the demand for control reserve does not increase if the *dynamic method* is used in the 100% renewable energy sources scenario (s. Table 2). If the *static method* is used, the control reserve approximately doubles. Another interesting result is that the maximum total control reserve for the *dynamic method* is almost the same as the total control reserve for the *static method*. The minimum total control reserve for the *dynamic method* equals one-third of the total control reserve in 2012.

Other studies showed the nRMSE of the wind and the photovoltaic forecast error is decisive for control reserve if a static method is applied [15,26]. Therefore a sensitivity analysis was carried out for the forecast error for the *dynamic method*. Nine scenarios were defined with three different values for the nRMSE of the 1h-forecast

Table 2
Comparison of the control reserve for the 100% renewable energy sources (RES) scenario obtained by the *static* and *dynamic methods* with the control reserve in 2012 in Germany.

	Static (100% RES)		Dynamic (100% RES)		2012 (Germany)	
	Pos. (MW)	Neg. (MW)	Pos. (MW)	Neg. (MW)	Pos. (MW)	Neg. (MW)
Ø Total reserve	11,149	8674	4488	4490	4535	4562
Min. total reserve	–	–	1650	1347	–	–
Max. total reserve	–	–	9870	7702	–	–
Ø Secondary reserve	4593	3848	2031	2022	2109	2149
Ø Tertiary reserve	6556	4826	2457	2468	2426	2413

Table 3Results of the sensitivity analysis for the nRMSE of the wind and photovoltaic 1h-forecast error for the *dynamic method*.

		nRMSE wind forecast error					
		0.69%		0.78%		0.86%	
		Pos. (MW)	Neg. (MW)	Pos. (MW)	Neg. (MW)	Pos. (MW)	Neg. (MW)
nRMSE	0.89%	4109	4087	4267	4465	4714	4807
photovoltaic	1.01%	4172	4113	4488	4490	4774	4831
forecast error	1.65%	4542	4295	4853	4663	5134	4998

error of wind and photovoltaic. The maximum nRMSE for the wind 1h-forecast error is 0.86% which equals the nRMSE assumed in the dena 2 study for 2020 [14]. The minimum nRMSE equals 0.69% which is 20% better than the maximum nRMSE. The middle nRMSE is 0.78% which is the nRMSE of the reference scenario. For the photovoltaic forecast error a maximum nRMSE of 1.65% is assumed which is the nRMSE of the persistence model developed at the Fraunhofer IWES. The minimum nRMSE is 0.89% which is 30% worse than the minimum forecast error of the wind energy. The middle nRMSE of 1.01% is also 30% worse than the middle nRMSE of wind forecast error. The photovoltaic forecast error is assumed to be worse than the wind forecast error since it is more complicated to forecast the power feed-in of photovoltaic [26].

The results (Table 3) show, that the positive total control reserve increases by 14.4% and the negative total control reserve increases by 11.3% in the scenarios with maximum nRMSE compared to the reference scenario. Another interesting result is that the total control reserve decreases by 8.4% (positive) and by 9.0% (negative) in the scenario with minimum nRMSE. The sensitivity analysis for the *dynamic method* shows that the total control reserve in a 100% renewable energy sources scenario does not increase drastically even if there is almost no improvement of the 1h-forecast.

5. Conclusions

Due to the increasing capacity of renewable energy sources, especially that of wind energy and photovoltaics, the amount of control reserve required will increase in the future if the same method is applied to size the control reserve. Thus, in this work, a new *dynamic method* for sizing the control reserve was developed. In contrast to the currently used *Graf-Haubrich method*, the *dynamic method* uses forecasts of the power feed-in of wind and photovoltaic energy for the period for which the control reserve is sized. The advantage is that if, for example, nominal power is predicted for wind energy, no negative control reserve is provided for wind energy. The *dynamic method* was applied to a 100% renewable energy sources scenario adopted from the Kombikraftwerk 2 project and the results were compared with those obtained using the *static method* for the same scenario. The *static method* yields results similar to those of the *Graf-Haubrich method*, but in contrast to this method, the *static method* can be applied to future scenarios. The *dynamic method* was used to size the control reserve one day ahead and for a product length of 1 h.

The result is that the average amount of total control reserve can be halved by using the *dynamic method* compared to the average amount obtained using the *static method*. Moreover, the maximum total control reserve for the *dynamic method* is less than the total control reserve for the *static method*. Another interesting result is that the average total control reserve for the *dynamic method* and the 100% renewable energy sources scenario equals the total control reserve currently needed in Germany.

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